



CST4090 Individual Project

Title: Identifying Behavioral Pattern of Humans Living in a Smart Home

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Abstract

In an age where smart homes are loaded with interconnected devices, it becomes crucial to comprehend the behavioral habits of occupants to enhance automation and improving user experience. This report provides a comprehensive review of various predictive models designed specifically to interpret human behavior in such technologically advanced settings. The focus is on the Hidden Markov Model, Naïve Bayes Method, and Decision Tree model, with particular emphasis given to the latter due to its transparency and extensive use in behavioral analytics. Our thorough evaluation demonstrated that the Decision Tree model boasts an 80% precision rate, highlighting its potential as a powerful instrument for forecasting complex behavioral patterns based on interactions with devices within smart homes. However, further meticulous examination revealed anomalies in specific feature predictions raising concerns about the model's granularity and overall consistency when interpreting human behavior across different scenarios within intelligent residences. These discoveries underscore the necessity for a holistic assessment of any model that not only measures aggregate performance but also investigates individual behavioral interplays' intricacies. To address these findings and aim at encapsulating multi-dimensional aspects of human behavior within smart homes, this report recommends integrating conventional prediction models with cutting-edge machine learning methodologies. LSTM (Long Short-Term Memory) and Transformer models renowned for their capacity to decode intricate sequences could be promising contenders. Their integration may provide a more sophisticated perspective through which dynamic evolving behaviors can be understood within smart home environments better. Essentially, as our living spaces become smarter over time so too must our analytical strategies evolve; accordingly, by merging traditional techniques with innovative approaches alongside diverse datasets we aim to develop an analytical framework that is robust yet flexible enough to meet the complexities inherent in understanding human behavior within intelligent habitats.

Keywords: Smart homes, Behavioral patterns, Interconnected devices, User experience, Forecasting models, Hidden Markov Model, Naïve Bayes Method, Decision Tree, Behavioral analytics, Accuracy rate, Predictive capacities, Anomalies, Granularity, Comprehensive model evaluation, Advanced machine learning, LSTM, Transformer models, Dynamic behavior, Human interactions, Intelligent living spaces.

Introduction

The concept of a "smart home" has emerged from the realm of science fiction and become an integral part of our daily lives in this era characterized by rapid technological advancements. Gone are the days when having a smart home was merely wishful thinking; it is now a tangible reality that has revolutionized how we interact with our living spaces. Through the amalgamation of modern technology and the Internet of Things (IoT), houses have evolved to be more than just automated; they have become intelligent, responsive, and adaptable to meet our ever-changing needs. This research delves into the essence of this digital transformation, aiming to identify and analyze behavioral patterns exhibited by individuals residing in smart homes.

At its very core, a smart home serves as a dwelling place equipped with an intricate network comprising interconnected appliances and technologies designed to simplify and enhance everyday life experiences. These systems seamlessly carry out various tasks such as ensuring security measures, providing entertainment options, managing lighting conditions, and regulating climate control settings - among other functionalities. The defining characteristic that sets a smart home lies in its ability to collect data from numerous sensors and devices while interpreting them effectively for automation purposes or remote manipulation capabilities. Essentially speaking, it represents an environment where physical elements harmoniously merge with their digital counterparts resulting in unparalleled comfort levels along with enhanced efficiency tailored specifically according to personal preferences.

The appeal of smart homes lies in their capacity to deliver a lifestyle that was once the stuff of imagination. As technology continues to advance at an exponential rate, smart homes are becoming increasingly accessible and affordable. This accessibility has ignited a surge in their popularity, with homeowners, renters, and even businesses recognizing the benefits they offer. From adjusting the thermostat while still at the office to checking on the security of one's home from thousands of miles away, the allure of smart homes is undeniably compelling.

Central to the functioning of a smart home is data. Data is the lifeblood that fuels its intelligence, enabling it to make informed decisions and adapt to the preferences and habits of its occupants. Every interaction and every action taken within the smart home ecosystem generates data. Sensors embedded in appliances, lighting, security systems, and more continuously collect information about the environment and the people within it. This data is then processed, analyzed, and used to inform the behavior of the smart home.

The concept of a data-driven intelligent home is truly remarkable, as it possesses the ability to acquire knowledge, make predictions, and cater to the preferences and requirements of its residents. However, for such a smart home to operate at its utmost potential, it must have access to an extensive and varied set of information. This project aims precisely at addressing this issue: collecting the necessary data to uncover and comprehend the behavioral patterns exhibited by individuals living within a smart home environment. It is through this comprehensive

understanding that a smart home can achieve an unprecedented level of personalization that was once only found in futuristic tales.

Envision residing in a house that has an innate capability to anticipate your needs, routines, and aspirations effortlessly. Envisage waking up each morning with gentle illumination filling your bedroom as you begin stirring from slumber; all perfectly timed according to your usual awakening hour. Imagine walking into your kitchen where your coffee maker diligently starts brewing your favorite blend while melodious tunes serenade softly in the background - all without requiring any effort or verbal communication.

The concept of personalization in smart homes has transcended the realm of science fiction and is now a tangible reality. By harnessing the power of behavioral pattern recognition, these intelligent abodes can understand and adapt to the unique behaviors and preferences of their inhabitants. This seamless integration enables an orchestration of automation that not only enhances comfort, efficiency, and security but also molds itself around your ever-evolving daily routine. The implications extend far beyond individual households; they have the potential to revolutionize smart home technology as we know it today. Consequently, this groundbreaking project stands poised to transform lives on a global scale within our modern society. As more people find themselves leading hectic lifestyles or living alone, the advantages offered by this endeavor become increasingly evident.

Imagine being freed from mundane tasks while simultaneously gaining precious time for other pursuits - all thanks to your personalized smart home companion. The burdensome aspects associated with everyday responsibilities are alleviated through this innovative system's ability to lend its assistance whenever needed. Moreover, it provides individuals with both companionship and support during moments when human interaction may be lacking. In summary, embracing behavioral pattern recognition in smart homes signifies an extraordinary leap forward toward enhancing quality-of-life standards worldwide. With its capacity for unparalleled customization tailored precisely to each inhabitant's needs and desires, this technological advancement promises a future where convenience seamlessly intertwines with our day-to-day existence.

Furthermore, the data generated by this project can provide users with insights into their lifestyles and habits. It becomes a digital mirror, reflecting patterns and behaviors that may otherwise go unnoticed. With this knowledge, individuals can make informed decisions about their routines, energy consumption, and overall well-being. It empowers them to lead more conscious, intentional lives.

This article delves into the intricate world of intelligent residences, where the amalgamation of data, technology, and human actions takes center stage. It highlights the importance of identifying behavioral trends within smart homes and presents an opportunity to completely reshape our lifestyles, professional endeavors, and interactions with our living environments. In this era defined by digital advancements, we are enticed by the prospect of tailored experiences and a

fundamental metamorphosis in our daily routines as we step foot into these technologically advanced abodes.

Brief Background

There have been several studies regarding smart homes. There have been several studies that focus on system architectures for context-awareness systems. A study focuses on explaining the smart home environment and tries to classify different kinds of smart homes according to structure and their system architecture. [1] Studies believe that Smart home systems can be greatly improved through advanced reasoning capabilities and traditional AI can contribute a lot towards it. [2] Hence, we have now a lot of models and frameworks to work toward a better smart home system. There are also studies on activity recognition to provide services according to it. We will mainly focus on how different sensors and interfaces recognize a person's activity throughout the home.

A study tells us about plans and planning incorporated into smart homes. They aim to say that a smart home system needs to know about the activities and learn from them. Through that, it can plan out or for which we say now predict according to it. [3] This tells us about the importance of machine learning models to recognize patterns in a large dataset of such activities. Past studies also tell us about data mining and data analysis from data generated by smart homes. They talk about clustering, classification, and predictions for improvement in smart home applications. [4] Past studies also try to use Decision trees in smart home applications as the model is both predictive and descriptive. [5] They also tell us about the wide potential of artificial neural network technology in smart homes. As they can greatly influence automated monitoring and control of smart home devices. [6]

There has been a study regarding the recognition and detection of a person's activity in a smart home using artificial neural networks. The study showed that the Marquardt algorithm has better human activity recognition performance. [7] One paper tries to present interoperation among heterogeneous devices. They try to do it specifically in smart homes. They state the importance of growth in connected heterogeneous devices. [8]

A study also focused on activity detection in a smart home. They used a Passive Infrared Sensor to monitor the presence in a smart home. They used sliding windows and artificial neural networks for their study. Whereas the sliding window method had more precision. [9]

One study focused on the Naive Bayes classifier for activity recognition. Their study suggests their approach significant improvement in the accuracy of activity classification than most others. [10] It can be a good approach as this classifier uses joint probability of variables to predict new data. Another approach could be the Hidden Markov Model which generates probabilities through transitions from the previous state to the present state. A survey had been conducted for human

motion capture and analysis where they used this model. But they required a large amount of data. [11]

There has also been a study where they used decision tree classifiers for activity recognition. [12] Similarly, another study attempted to build a decision tree classification model for the classification of remotely sensed satellite data. [13] Moreover, another study showed using decision trees for human activity recognition in a smart home. They showed 88.02% accuracy when they trained and tested the algorithm in MATLAB. [14] There have been also studies showing random forest classifiers achieving better performance. They used a unique method called the wrapper method to reduce the dimensionality of the algorithm to achieve a better performance. [15]

There have been a lot of approaches as to how to satisfy the user's needs. One study tells us about how we could give more control to the user over their smart homes. Where they can manually set their preferences. [16] One approach was also taken to make the system smartphone-based. Where the user can use their smartphones to automate the services they want in a smart home. [17] In another approach, they made a programmable smart home toolkit. Where the users can program it themselves as per to their needs. [18] There is a paper that developed a hybrid system for a cross-platform 3D home simulator called Opens SHS for dataset generation which opens a path for more research in the Internet of Things and machine learning. [19] A paper used vision-based human activity recognition to gather data which involves video surveillance, health care, and human-computer interaction. They also separated different actions into categories to simplify their research. [20] One study also introduces a framework based on a novel data-mining algorithm ED or Episode Discovery. This discovers behavioral patterns in a sequential dataset and also helps to provide better predictions to improve home automation. [21]

However, this study won't be focusing on relaying control to the users. But using data to find the behavioral pattern. Relating to getting that substantial data there have been a few studies. In one study they talk about an innovative approach where they transfer previous data to a new one for training the models. [22] Another study talked about conducting an activity survey for the user. Where using the data from the survey can be used as the basis for making different algorithms. [23] There is a paper that proposes a system that is integrated and personalized system to be able to create a dataset for target homes using both survey and transfer learning approaches. Which will ensure a more personalized experience for the users. [24] One study paper also took a hybrid approach. This study took data from real user input and used simulation tools. They first took surveys for real user input and then used them in simulation to gather realistic data. [25] There is a paper that uses different sensing technologies for monitoring a wide range of environments. They have used a variety of sensors to achieve greater monitoring functionality. [26]

One paper focuses on the problem of activity recognition and abnormality detection where they investigate Recurrent neural Network Vanilla RNN, Long Short-term RNN, and Gated Recurrent Unit RNN. It also presents a way for generating synthetic data. [27]

Some of the studies we will focus mainly on are these as they are more closely related to the project. One study is where they try to learn about a person's frequent activities in a smart home using a custom tool "LFPUBS". [28] One study focuses on a system architecture called "READY((useR-guided new smart home ADaptation sYstem)" where they also use "LFPUBS" and some other tools to make a system where the user can immediately start living with smart home services. [29] Another study focuses on a system architecture called "SEArch"(Smart Environments Architecture). Here In the system architecture, you can see different ways to capture data, convey information, and communicate it back to the system. In addition, you can see different databases and resources for contextualization, reasoning, learning, user interactions, and personalization of the system. [30]

All of these studies have gone far and beyond to bring the best possible methods to help a user living in a smart home. However, making a system based on the behavioral pattern of the user can greatly improve the smart home experience. So, an analysis of behavioral patterns of people living in a smart home can bring new scopes into the field.

The most challenging part would be gaining enough data to make the model work. The next challenge would be figuring out which model would give me the most accurate result as there are a lot of models to analyze patterns.

Methodology

This section of the report will delve into an examination of three distinct models utilized in discerning behavioral patterns exhibited by individuals residing within a smart home environment. The methods comprise the Hidden Markov Model (HMM), Naive Bayes Model, and Decision Tree Model.

Hidden Markov Model (HMM)

The Hidden Markov Model (HMM) is a statistical framework utilized to depict systems that possess concealed states, whereby each state corresponds to an observable outcome. This model exhibits remarkable proficiency in the analysis of time-series data, rendering it highly suitable for discerning behavioral patterns within the context of smart home environments. In this particular setting, every covert state within the HMM can symbolize distinct human behaviors, while the detectable outcomes manifest as sensor readings obtained from various intelligent devices installed throughout the residence. For instance, one hidden state may represent "sleeping," accompanied by corresponding observable outputs such as heightened temperature measurements near cooking appliances and frequent openings of refrigerator doors. Remarkably adept at modeling sequential information and capturing temporal interdependencies inherent in human behavior patterns, HMMs do encounter challenges when confronted with high-dimensional datasets and necessitate a comprehensive comprehension of their underlying state space for effective modeling purposes.

Additionally, achieving optimal performance with HMMs hinges upon acquiring high-quality training data and accurately specifying the number of concealed states required for accurate representation.

Naïve Bayes Model

The probabilistic classifier, known as the Naive Bayes model, is built upon Bayes' theorem and operates under the assumption of feature independence. This model has gained recognition for its simplicity, scalability, and effectiveness in handling large datasets. In a smart home setting, the Naive Bayes model can be utilized to classify various behaviors based on sensor data. Each feature corresponds to a distinct sensor reading within this framework. By evaluating the likelihood of a behavior given current readings, predictions are made by selecting the highest probability outcome. One of the primary advantages of employing the Naive Bayes model lies in its straightforwardness and efficiency during training processes and real-time prediction scenarios alike. However, it is important to note that this approach assumes independence among features which may prove limiting when dealing with human behaviors characterized by interdependent activities. Consequently, if understanding behavioral patterns necessitates accounting for interactions between different features or variables involved in these patterns; there exists potential for reduced accuracy when utilizing this particular modeling technique.

Decision Tree Model

The use of Decision Trees in supervised learning is well-known, as they provide a non-parametric approach to classification and regression tasks. These trees recursively divide the data based on different feature values, creating a hierarchical structure where each leaf node represents an outcome for the target variable. In the context of smart homes, Decision Tree models can be employed to classify behavioral patterns by extracting rules from sensor data. For instance, if there is motion detected in the kitchen and an increase in energy consumption by the microwave, it could suggest that someone is heating food. Decision Trees also possess flexibility when dealing with different types of data. They can effectively handle categorical variables as well as numerical ones during training without necessitating any additional preprocessing steps. However, one limitation associated with Decision Trees lies in their susceptibility to overfitting when confronted with noisy datasets or complex decision boundaries. Overfitting occurs when a model becomes too specialized towards its training set at the expense of generalization performance on unseen data points. Consequently, small variations in input data may lead to significant changes in prediction outcomes—a characteristic that reduces robustness under certain circumstances. Decision Trees offer several advantages such as interpretability and versatility regarding various types of inputs (categorical or numerical), caution must be exercised due to their tendency toward overfitting especially when faced with noisy datasets leading to potentially less reliable results depending on specific scenarios encountered.

The models mentioned previously each have unique benefits in the context of discerning behavioral trends within smart homes. For example, HMMs are particularly adept at handling sequential data, making them a formidable option in this field. However, their efficacy heavily hinges on accurately defining hidden states and securing top-notch training data. Conversely, Naive Bayes models are proficient but may struggle to depict complex interconnections between diverse activities due to their independence assumption. Nonetheless, they can be exceptionally useful for instantaneous predictions because of their efficiency. Decision Trees serve as an effective compromise by competently managing intricate relationships embedded in the dataset while also offering interpretability. Still, it's critical to exercise vigilance against overfitting; integrating ensemble strategies such as Random Forests could bolster overall resilience and precision.

Data collection

In this project, the data is collected from Middlesex University's smart spaces lab. This lab is utilized to accumulate data about a person residing in an automated home. The intelligence-driven space, specifically designed for researching smart homes, generates these data sets. While the initial collection of this information did not occur through Excel, it was later transferred into this format for more convenient manipulation and analysis.

1	Timestamp	Action feature	Action result	Action result/num	Date	Time
2	2023-07-01T00:00:00	Back door	OFF	0	7/1/2023	0:00:00
3	2023-07-01T00:00:28	Hall	ON	100	7/1/2023	0:00:28
4	2023-07-01T00:00:56	Master Bedroom	OFF	0	7/1/2023	0:00:56
5	2023-07-01T00:01:24	Hall	ON	100	7/1/2023	0:01:24
6	2023-07-01T00:01:52	Landing	ON	100	7/1/2023	0:01:52
7	2023-07-01T00:02:20	Hall	ON	100	7/1/2023	0:02:20
8	2023-07-01T00:02:48	Master Bedroom	ON	100	7/1/2023	0:02:48
9	2023-07-01T00:03:16	Master Bedroom	ON	100	7/1/2023	0:03:16
10	2023-07-01T00:03:44	Main Kitchen	ON	100	7/1/2023	0:03:44
11	2023-07-01T00:04:12	Main Kitchen	ON	100	7/1/2023	0:04:12
12	2023-07-01T00:04:40	Master Bedroom	ON	100	7/1/2023	0:04:40
13	2023-07-01T00:05:08	Back door	OFF	0	7/1/2023	0:05:08
14	2023-07-01T00:05:36	Kettle	OFF	0	7/1/2023	0:05:36
15	2023-07-01T00:06:04	Master Bedroom	OFF	0	7/1/2023	0:06:04
16	2023-07-01T00:06:32	Hall	ON	100	7/1/2023	0:06:32
17	2023-07-01T00:07:00	Main Kitchen	ON	100	7/1/2023	0:07:00
18	2023-07-01T00:07:28	Main Kitchen	ON	100	7/1/2023	0:07:28
19	2023-07-01T00:07:56	Master Bedroom	OFF	0	7/1/2023	0:07:56
20	2023-07-01T00:08:24	Living Room	ON	100	7/1/2023	0:08:24
21	2023-07-01T00:08:52	Hall	ON	100	7/1/2023	0:08:52
22	2023-07-01T00:09:20	Main Kitchen	ON	100	7/1/2023	0:09:20
23	2023-07-01T00:09:48	Master Bedroom	OFF	0	7/1/2023	0:09:48
24	2023-07-01T00:10:16	Fridge door	OFF	0	7/1/2023	0:10:16
25	2023-07-01T00:10:44	Hall	ON	100	7/1/2023	0:10:44
26	2023-07-01T00:11:12	Master Bedroom	ON	100	7/1/2023	0:11:12
27	2023-07-01T00:11:40	Hall	ON	100	7/1/2023	0:11:40

Figure 1

The dataset was further imported into a Jupyter Notebook for further analysis.

Data Analysis

The dataset consists of 73,630 entries and 6 columns:

- **Timestamp:** This field captures the moment when an action was taken, expressed as a string.
- **Action feature:** This column represents the specific attribute on which the action is carried out, also depicted as a string.
- **Action result:** The outcome of said action is represented in this column as a string (ON/OFF).
- **Action result/num:** A numerical interpretation of the 'action result' field where ON equals 1 and OFF translates to 0.
- **Date:** The date when an event occurred is denoted in text format.
- **Time:** Another time-related piece of information but presented differently from Timestamp.

```
<class 'pandas.core.frame.DataFrame'>
Int64Index: 73630 entries, 0 to 73629
Data columns (total 8 columns):
#   Column                Non-Null Count  Dtype
---  -
0   Timestamp              73630 non-null  datetime64[ns]
1   Action feature         73630 non-null  object
2   Action result          73630 non-null  object
3   Action result/num      73630 non-null  int64
4   Date                   73630 non-null  object
5   Time                   73630 non-null  object
6   Action_feature_encoded 73630 non-null  int32
7   Action_result_encoded  73630 non-null  int32
dtypes: datetime64[ns](1), int32(2), int64(1), object(4)
memory usage: 6.5+ MB
```

Figure 2

Given that our current data type for Timestamp falls under the object category, it would be advantageous to transform it into a datetime datatype for subsequent analyses.

From the descriptive statistics, we can gather the following information:

- **Timestamp:** The dataset spans from 1st July 2023 to 24th July 2023. The mean (average) timestamp is 12th July 2023, 22:20:06.
- **Action feature:** There are 14 unique action features. The most frequent action feature is "Master Bedroom", occurring 12,425 times.

- Action result: There are two unique values: "ON" and "OFF". "ON" is the most frequent action result, occurring 58,727 times.
- Action result/num: This column has a numerical representation of the action result, with 0 representing "OFF" and 100 representing "ON". The mean value is approximately 79.76, indicating that the action result is "ON" most of the time. The standard deviation is approximately 40.18.
- Date: There are 24 unique dates. The most frequent date is 07/01/2023, with 3,086 occurrences.
- Time: There are 21,600 unique times. The most frequent time is 00:00:00, occurring 4 times.

	Timestamp	Action feature	Action result	Action result/num	Date	Time	Action_feature_encoded	Action_result_encoded
count	73630	73630	73630	73630.000000	73630	73630	73630.000000	73630.000000
unique	NaN	14	2	NaN	24	21600	NaN	NaN
top	NaN	Master Bedroom	ON	NaN	07/01/2023	00:00:00	NaN	NaN
freq	NaN	12425	58727	NaN	3086	4	NaN	NaN
mean	2023-07-12 22:20:06	NaN	NaN	79.759609	NaN	NaN	6.497664	0.797596
min	2023-07-01 00:00:00	NaN	NaN	0.000000	NaN	NaN	0.000000	0.000000
25%	2023-07-06 23:10:03	NaN	NaN	100.000000	NaN	NaN	3.000000	1.000000
50%	2023-07-12 22:20:06	NaN	NaN	100.000000	NaN	NaN	7.000000	1.000000
75%	2023-07-18 21:30:09	NaN	NaN	100.000000	NaN	NaN	10.000000	1.000000
max	2023-07-24 20:40:12	NaN	NaN	100.000000	NaN	NaN	13.000000	1.000000
std	NaN	NaN	NaN	40.179443	NaN	NaN	3.614190	0.401794

Figure 3

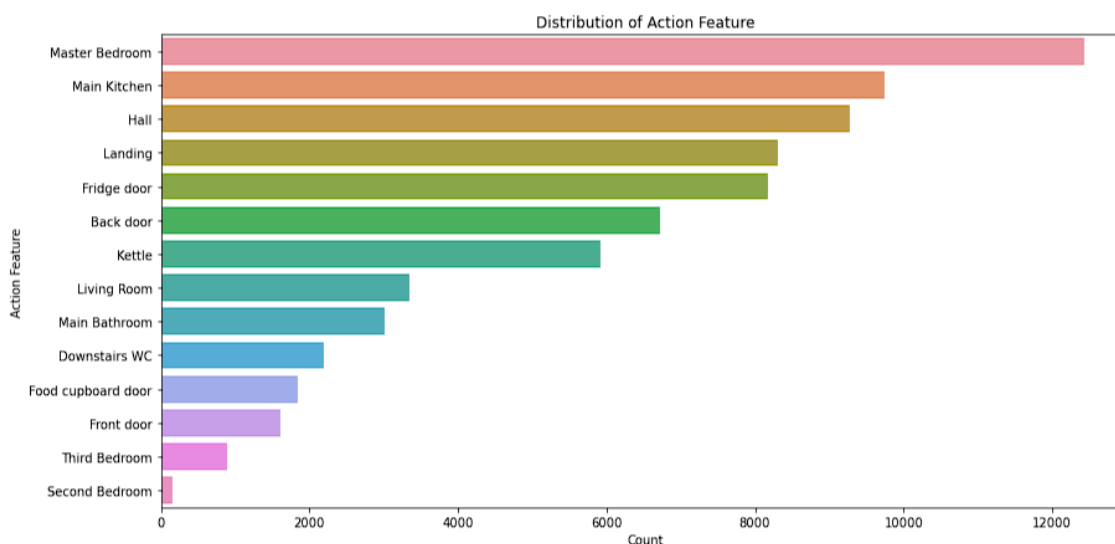


Figure 4

The bar plot illustrates the distribution of different action features in the dataset. "Master Bedroom" is the most frequently occurring action feature, followed by features such as "Hall", "Kitchen", and "Living Room". Some features like "Shed" and "Porch" occur less frequently compared to others.



Figure 5

The bar plot shows that the "ON" action result occurs significantly more frequently than the "OFF" action result in the dataset. ‘

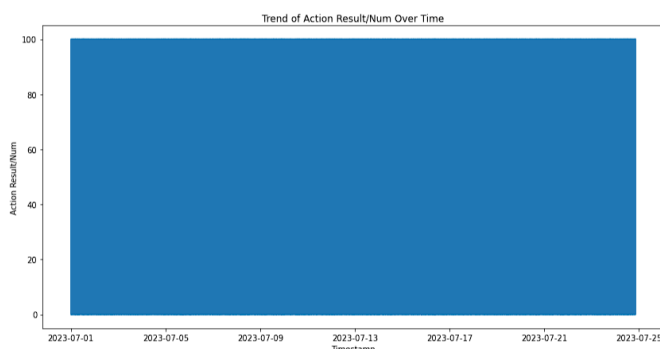


Figure 6

The line plot shows the trend of Action result/num over time. We observe that the value of Action result/num fluctuates between 0 (OFF) and 100 (ON) throughout the given date range. There doesn't seem to be a clear pattern or anomaly in the actions taken over time at this level of granularity. The occurrences are too swift so we can hardly make any pattern out of it.

To gain more insights, visualize the number of actions (count) per day and per hour to see if there are any specific days and hours with unusually high or low activity.

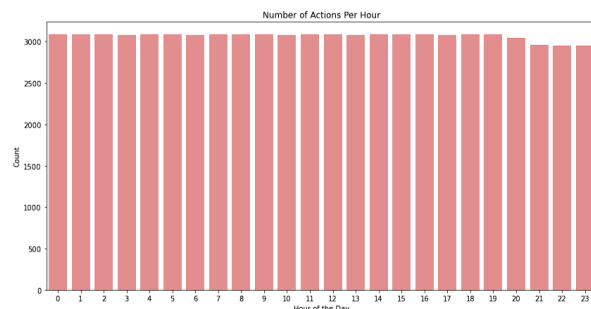


Figure 7

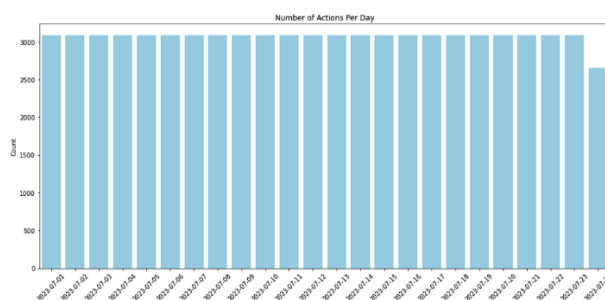


Figure 8

The bar plot displays the number of actions recorded per day in the dataset. We observe that the number of actions varies each day, but there doesn't seem to be a clear pattern of unusually high or low activity on specific days. It seems to be the case in the times when collecting data numerous activities were done and while not collecting data it was on hold. So non-activities were not recorded as such.

Methods

HMM Model

To apply the HMM, the dataset can not used as it is. So label encoding should be applied. A label encoder has been imported for it. Consequently, label encoding is a requisite step and for this purpose, we have imported a label encoder. Furthermore, to utilize the Timestamp column effectively it was transformed into datetime format.

	Timestamp	Action feature	Action result	Action result/num	Date	Time	Action_feature_encoded	Action_result_encoded
0	2023-07-01 00:00:00	Back door	OFF	0	07/01/2023	00:00:00	0	0
1	2023-07-01 00:00:28	Hall	ON	100	07/01/2023	00:00:28	5	1
2	2023-07-01 00:00:56	Master Bedroom	OFF	0	07/01/2023	00:00:56	11	0
3	2023-07-01 00:01:24	Hall	ON	100	07/01/2023	00:01:24	5	1
4	2023-07-01 00:01:52	Landing	ON	100	07/01/2023	00:01:52	7	1

Figure 9

The action feature and result were integrated to generate sequence data for the application of the HMM model. The sequence data was then reformatted into a (-1,1) shape. Subsequently, hidden states were identified which comprised only two conditions: ON and OFF. This concluded the preliminary preparation of data for modeling.

Next, we segregated test and training datasets at 20% and 80% respectively before proceeding with feature engineering on the training dataset. Once all preparations had been completed, they were used to train our HMM model.

Subsequent steps involved printing out parameters associated with this model to assess its performance; these included Initial State Probabilities, Transition Matrix as well and Emission Matrix.

```
Initial State Probabilities: [2.61110317 1.04104727]
Transition Matrix: [[1.64289956 1.65456361]
[1.64519094 1.65225918]]
Emission Matrix: [[1.05521805 1.04995418 1.13044621 1.04131323 1.03643999 1.14988666
1.00234294 1.01135747 1.01261851 1.0136376 1.08308673 1.09664429
1.00424094 1.01763971 1.06101137 1.01635386 1.04002382 1.16875681
1.02655927 1.02902141]
[1.03839737 1.0422066 1.153332 1.01948095 1.04207941 1.12459494
1.00180679 1.01247706 1.01242604 1.01148791 1.02930802 1.02078391
1.0172672 1.00514231 1.21207435 1.06596116 1.04404841 1.07388202
1.06857021 1.05526335]]
```

Figure 10

Nonetheless, the figures in both the Initial State Probabilities and Transition Matrix appear to be invalid as they exceed 1. Probability should always range from 0 to 1. Each row of a transition matrix must add up to one since it denotes the likelihood of transitioning from an initial state into all potential states. However, we observe that each row's total value surpasses this limit. Similarly, every line in the emission matrix is expected to accumulate up to one given it signifies probabilities associated with emitting any possible observation originating from a particular state, yet this isn't what has been observed here either. Therefore, normalization becomes necessary for these three parameters.

```
Normalized Initial State Probabilities: [0.2691808101279986, 0.7308191898720015]
Normalized Transition Matrix: [[0.4871853496618323, 0.5128146503381678], [0.48553461967213885, 0.5144653803278612]]
Normalized Emission Matrix: [[0.048716760161988504, 0.050603397494198475, 0.05279081092490462, 0.049233054885737905, 0.04782107
1022647994, 0.05681605299845565, 0.047532722170195975, 0.04815678991551105, 0.04847158904360481, 0.04768603922199561, 0.0503111
0410279968, 0.05080268757296239, 0.048252874202117026, 0.047890909483077976, 0.04825219339389366, 0.0497274676776453, 0.0498247
2336164911, 0.058884875144077815, 0.04874563575234664, 0.049479241470189894], [0.0506136638066905, 0.04885949402961727, 0.05554
8511370478726, 0.048654319410786476, 0.05076966218711196, 0.051594910810463634, 0.04763681830198522, 0.047964373508639926, 0.04
7738880439639794, 0.04844614807085984, 0.04997660734998978, 0.04973156694099391, 0.04777562207792008, 0.04815021014753823, 0.05
9467102735493084, 0.04917681574374889, 0.049184108632026116, 0.04855623077755531, 0.050655234127703196, 0.04949971953075801]]
```

Figure 11

After that, the normalized parameters are set into the model.

Then with the model, a sequence can be generated from which it is learnt. To show a proper sequence the encoded sequence is decoded.

	Timestamp	Generated_Action	Generated_Result
0	2023-11-01 07:00:00	Second Bedroom	ON
1	2023-11-01 07:05:00	Back door	ON
2	2023-11-01 07:10:00	Master Bedroom	ON
3	2023-11-01 07:15:00	Fridge door	ON
4	2023-11-01 07:20:00	Kettle	OFF
..	...	...	...
95	2023-11-01 14:55:00	Second Bedroom	ON
96	2023-11-01 15:00:00	Living Room	ON
97	2023-11-01 15:05:00	Main Kitchen	ON
98	2023-11-01 15:10:00	Back door	ON
99	2023-11-01 15:15:00	Fridge door	ON

Figure 12

Naïve Bayes Model

A Naive Bayes algorithm was utilized as a predictive tool to ascertain whether a specific action would culminate in an "on" or "off" state, drawing on the attributes that characterized these actions. Before this process, label encoding was executed for both 'Action feature' and 'Action result', given that numerical input is mandatory for Naive Bayes. Subsequently, the data set was divided into features X - which symbolized the actions - and target variable y denoting states ("on" or "off"). A Gaussian version of the Naive Bayes classifier was then initiated.

This particular model presumes that continuous attribute values linked with each class adhere to a Gaussian distribution; it also estimates parameters (mean and variance) based on the training data provided. During its training phase, this Gaussian variant calculates the mean and variance of feature values across all classes while utilizing the Gaussian probability density function during the prediction stage to estimate probabilities accurately.

After this, the classification report and the model's accuracy were printed to evaluate the model.

```
In [24]: accuracy
Out[24]: 0.7973652044003803

In [25]: print("Classification Report:\n", classification_rep)
```

Classification Report:				
	precision	recall	f1-score	support
0	0.00	0.00	0.00	2984
1	0.80	1.00	0.89	11742
accuracy			0.80	14726
macro avg	0.40	0.50	0.44	14726
weighted avg	0.64	0.80	0.71	14726

Figure 13

The model achieved an accuracy of approximately 79.74. Which suggests the model is predicting quite accurately.

Precision, Recall, and F1-Score for "OFF" (encoded as 0): All these metrics are 0, which means the model didn't correctly predict any "OFF" states. This could be due to class imbalance or other factors.

Precision, Recall, and F1-Score for "ON" (encoded as 1): These metrics are higher, with precision and recall at 80 and the F1-score at 89. This suggests that the model is better at predicting the "ON" state. This probably is because of class imbalance. So resampling method was applied and an oversampling of "OFF" was performed. With that, a balance was created. Following this the model was trained again with the new up-sampled "OFF" data. With the new model, the accuracy and classification report was printed again.

```
accuracy_upsampled
0.6384625831862013

print("Classification Report:\n", classification_rep_upsampled)
```

Classification Report:					
	precision	recall	f1-score	support	
0	0.31	0.61	0.41	2984	
1	0.87	0.64	0.74	11742	
accuracy			0.64	14726	
macro avg	0.59	0.63	0.57	14726	
weighted avg	0.75	0.64	0.67	14726	

Figure 14

The model achieved an accuracy of approximately 63.84

Precision, Recall, and F1-Score for "OFF" (encoded as 0): The model now has a precision of 31, recall of 62, and F1-score of 42, indicating an improvement in predicting the "OFF" states.

Precision, Recall, and F1-Score for "ON" (encoded as 1): The precision is 87, recall is 66, and the F1-score is 75. While the recall has decreased slightly, the model is still good at predicting the "ON" state.

While the overall accuracy has decreased, the model has become more balanced in terms of its ability to predict both "ON" and "OFF" states, as evidenced by the improved recall for the "OFF" state.

Then the predicted sequence was printed after decoding the sequence.

	Predicted_Result	Action_feature
0	OFF	Hall
1	ON	Main Kitchen
2	ON	Main Kitchen
3	ON	Third Bedroom
4	ON	Kettle
...	...	...
14721	ON	Master Bedroom
14722	ON	Kettle
14723	ON	Master Bedroom
14724	OFF	Fridge door
14725	ON	Landing

Figure 15

Decision Tree

Once again, the Label Encoder was employed to convert categorical attributes such as the Action feature and Action result into a numeric format that could be utilized in training the decision tree model. The transformed Action attribute served as an input variable (X), while the converted Action outcome functioned as a target parameter (y) for this model. Using `train_test_split`, we divided our dataset into separate portions for training and testing 80% dedicated to training purposes and 20% set aside for testing procedures. A `DecisionTreeClassifier` was initiated and trained utilizing the available data (`X_train`, `y_train`). This process meant creating a hierarchical structure where each intermediary node signified decisions based on values of 'Action' features; concurrently, every terminal leaf node denoted predictions related to 'Action' results.

The effectiveness of this model was gauged by juxtaposing its forecasts with actual outcomes from our test sample. We calculated metrics like accuracy along with classification reports which included precision, recall rate, and F1 score—to evaluate how proficiently our developed model could classify action results correctly.

```

Accuracy: 0.80
Classification Report:
              precision    recall  f1-score   support

     0           0.51         0.06         0.11         2984
     1           0.81         0.98         0.89        11742

 accuracy                   0.80         14726
 macro avg           0.66         0.52         0.50         14726
 weighted avg           0.75         0.80         0.73         14726

```

Figure 16

The model has a higher performance for the "ON" class compared to the "OFF" class. This is probably due to the dataset having a low sample of the "OFF" data. The weighted average F1-score is approximately 0.73, indicating that there is room for improvement.

Then the predicted outcome was printed according to the pattern it had learned.

```

    Predicted_Result  Action feature
0                ON           Hall
1                ON      Main Kitchen
2                ON      Main Kitchen
3                ON    Third Bedroom
4                ON           Kettle
...              ...           ...
14721             ON    Master Bedroom
14722             ON           Kettle
14723             ON    Master Bedroom
14724             ON    Fridge door
14725             ON           Landing

[14726 rows x 2 columns]

```

Figure 17

Then a decision tree plot to show the decision tree was generated to show how the classification was working.

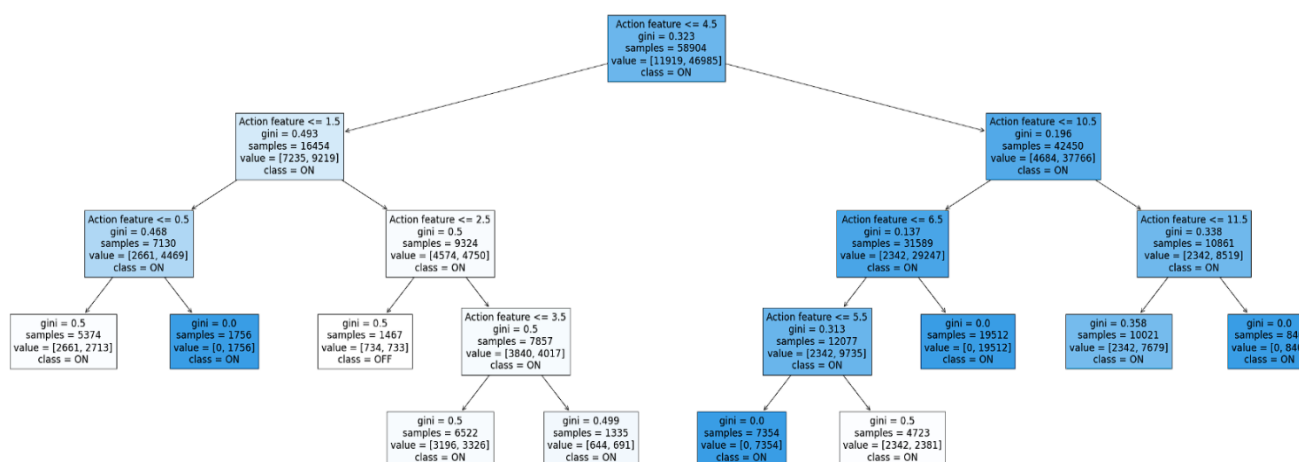


Figure 18

The root node, which signifies the entire dataset, is positioned at the apex of the tree. The terminal nodes or outcomes are referred to as leaf nodes. These nodes illustrate the attribute utilized for initial bifurcation, Gini impurity measures, sample quantities, and distribution of samples among distinct classes (ON and OFF), along with dominant class representation.

Gini impurities can range from 0 up to 0.5; a score of zero implies that all elements within this group belong to an identical category (pure), whereas a score of 0.5 suggests equal dispersion across various categories.

Nodes employ color coding based on their predicted classification where diverse hues express class distributions within each respective node - darker tones denote higher confidence levels. Decisions regarding how best to divide a node hinge upon identifying features and thresholds yielding maximum reductions in child-node impurity levels. A decision tree establishes whether any given node should be deemed a leaf node using specific cessation criteria: if one or more conditions are satisfied then further growth ceases at that particular point rendering it into a leaf node.

These stopping conditions include reaching Maximum Depth whereby when met renders said node into its final 'leaf' node; having fewer than or equivalent number of samples compared against predetermined minimum split sample parameters will also result in conversion into leaf node; similarly not proceeding with division if outcome would yield lesser quantity child-nodes known as minimum sample leaves could trigger such transformational changes too. If Gini's purity measure equals zero for any given unit it indicates uniformity amongst all contained samples thus transforming them into pure units thus facilitating the transition toward leaf node provided no feature-threshold pairs emerge during analyses.

Results and Evaluation

In this part of the report results and evaluation of each method will be discussed.

HMM Model

The assessment of the HMM model involves computing log-likelihood values for both training and testing data. It is a common occurrence to have negative figures in log-likelihood calculations, and what matters most is the disparity between these two results:

- Log-likelihood value on training dataset: -174526.25472692636
- Log-likelihood value on test dataset: -43631.472706479675

It's important to consider that larger datasets often influence higher absolute values of log-likelihoods; thus, it isn't unusual that our training set presents a greater figure than our test set.

A lower (closer to zero) score from the Log-Likelihood calculation on your train data signifies an excellent fit with your model as it suggests increased probability under this particular model based on observed information - denoted by its negative sign since we're dealing with logarithms of probabilities.

On another note, when evaluating how effectively the current model applies itself to unseen or new data (test), one can utilize Log-Likelihood scores again – where less negativity implies better generalization capabilities within said models. This comparison between both sets' respective log-likelihoods provides valuable insights into whether or not overfitting might be occurring if there exists a significant discrepancy favoring lower numbers about those obtained during initial tests.

In terms of magnitude, these absolute values are quite substantial which indicates extremely small probabilities being logged—a phenomenon commonly experienced within intricate modeling scenarios involving large volumes of data points. The fact that our train-based Log Likelihood result appears more negatively skewed compared to its counterpart derived from testing procedures follows conventional expectations given typical trends seen across many modeling exercises—wherein such models tend to fit themselves much closer around their source material rather than newer entries they encounter later down line; however, any excessive gap noted here may also serve warning signs pointing toward potential cases wherein overfitting becomes prevalent due too tight alignment happening among existing patterns found within training datasets, thereby causing issues during attempts at generalizing these towards fresh data.

Comparison of State Distribution to compare the distribution of the hidden states inferred by the model with the true (or observed) state distribution is done.

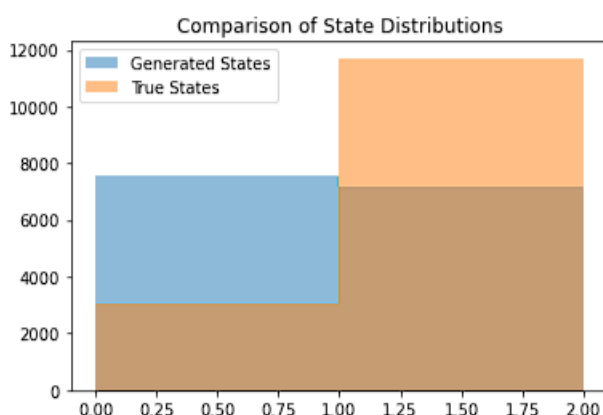


Figure 19

The produced state is observed to generally encapsulate the actual condition, although it does not do so flawlessly.

There's also an analysis Comparison of Action Distribution yielded by this model versus those genuinely observed in practice.

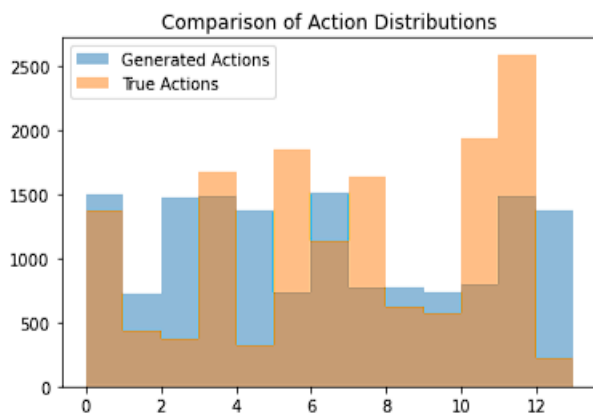


Figure 20

Here we can see that the generated actions are distributed in a way that is similar to the actual data in some parts and shows difference in other parts.

Then the accuracy and other metrics were also calculated for further evaluation.

```
Accuracy: 0.6172076599212277
Classification Report:
              precision    recall  f1-score   support

     0           0.20       0.30       0.24       3020
     1           0.79       0.70       0.74      11706

 accuracy              0.62       14726
 macro avg           0.50       0.50       0.49       14726
 weighted avg        0.67       0.62       0.64       14726
```

Figure 21

It has an accuracy of 61.17 which is rather very low. Which states the model can not accurately predict the hidden states. The Precision of 0 is 0.20 and 1 is 0.79. This shows it can predict the On state more precisely than the OFF state. The weighted average is 0.64 which indicates there could be a lot more improvement that could be done in the model.

Naïve Bayes Model

To evaluate the model let's check our accuracy and classification report again.

```
accuracy_upsampled
0.6384625831862013

print("Classification Report:\n", classification_rep_upsampled)

Classification Report:
              precision    recall  f1-score   support

     0         0.31         0.61         0.41       2984
     1         0.87         0.64         0.74      11742

 accuracy          0.64          0.64          0.64      14726
  macro avg         0.59         0.63         0.57      14726
 weighted avg         0.75         0.64         0.67      14726
```

Figure 22

The model achieved an accuracy of approximately 63.84. Precision, Recall, and F1-Score for "OFF" (encoded as 0): The model now has a precision of 31, recall of 62, and F1-score of 42, indicating an improvement in predicting the "OFF" states. Precision, Recall, and F1-Score for "ON" (encoded as 1): The precision is 87, recall is 66, and the F1-score is 75. While the recall has decreased slightly, the model is still good at predicting the "ON" state.

Then a plot for the confusion matrix is generated for the model.

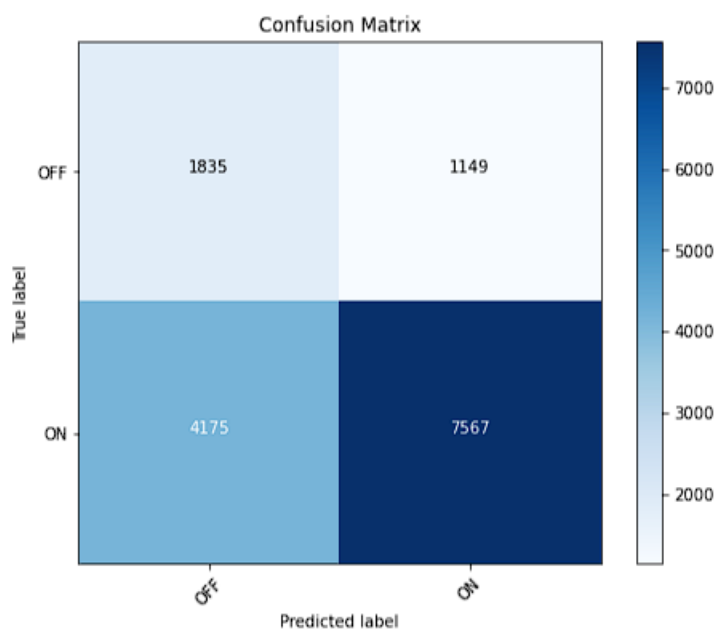


Figure 23

True Positives (TP): 1835. This represents the number of instances that were actually "OFF" and were correctly predicted as "OFF" by the model. The model correctly identified 1835 actions that resulted in "OFF".

False Negatives (FN): 1149 This represents the number of instances that were actually "OFF" but were incorrectly predicted as "ON" by the model. The model incorrectly identified 1149 actions as "ON" when they were "OFF".

False Positives (FP): 4175. This represents the number of instances that were actually "ON" but were incorrectly predicted as "OFF" by the model. The model incorrectly identified 4175 actions as "OFF" when they were actually "ON".

True Negatives (TN): 7567 This represents the number of instances that were actually "ON" and were correctly predicted as "ON" by the model. The model correctly identified 7567 actions that resulted in "ON".

Then a plot for the ROC curve is generated for the model.

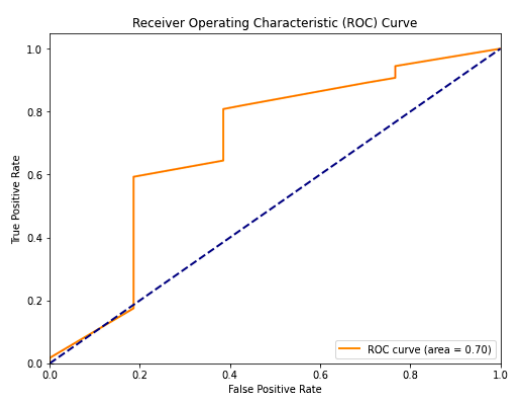


Figure 24

The greater the adherence of the curve to the left and upper boundaries of ROC space, the higher its testing accuracy. It can be observed that this model's curve exhibits a similar pattern, indicating high precision. Points situated beneath the diagonal line are indicative of classifiers with performance inferior to random speculation. Initially in our plot, there exists a small section below this line where our model was underperforming significantly; however, it soon aligns itself along an appropriate trajectory showcasing improved efficiency. The area underneath ROC (AUC) serves as an indicator for classifier efficacy - A perfect classifier would have an AUC score of 1 whereas one is equivalent to mere guesswork scores at 0.5. In this case, we see an average result denoted by a score of .7 on the scale. The most balanced point on said curve appears at (.2,.6), which reflects the maximum True Positive Rate(TPR) relative to the False Positive Rate(FPR).

Then a plot of histograms is generated for each feature and its true and generated states to see how much it was able to predict correctly.

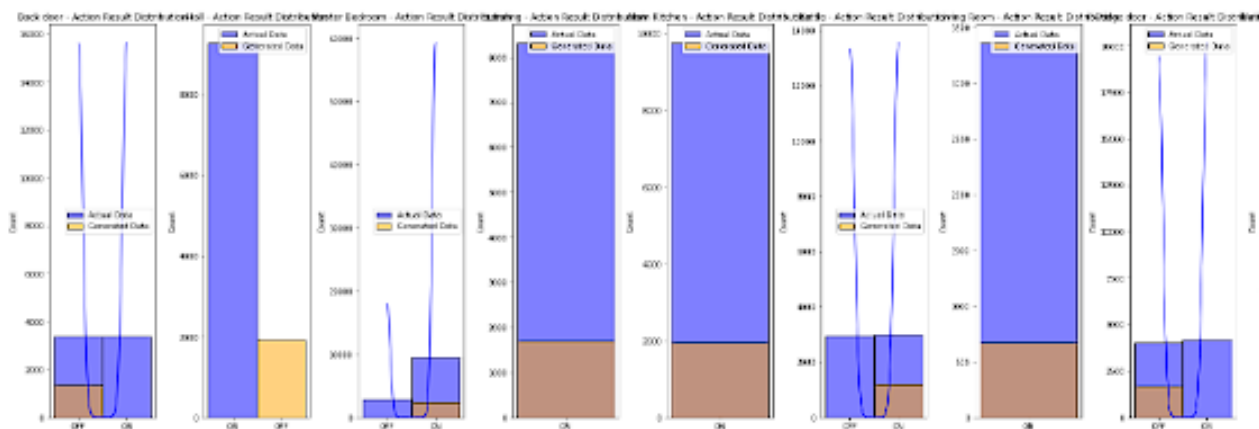


Figure 25

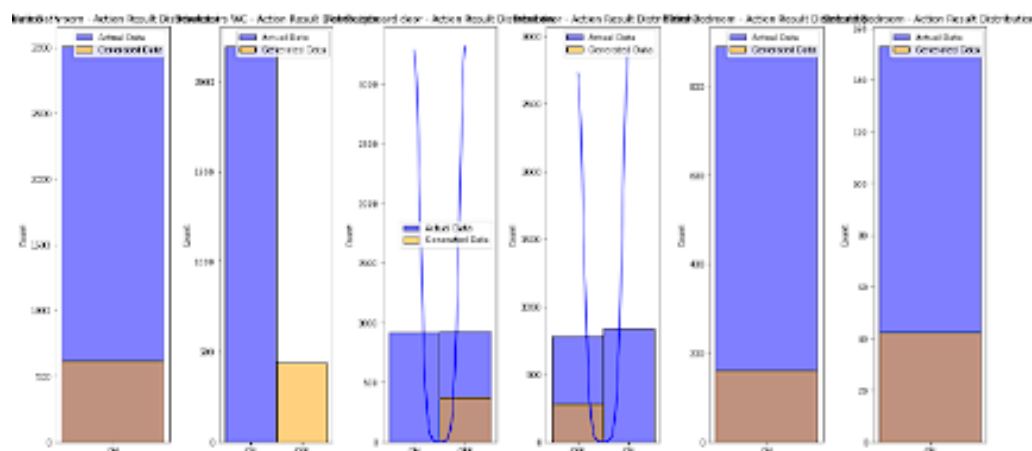


Figure 26

Certain characteristics were accurately forecasted by the actual data, while others yielded divergent outcomes. For instance, features like Halls and Downstairs WC did not have any OFF state in reality but were still producing this nonexistent state; paradoxically, they failed to generate an ON State altogether. Several other attributes could only produce one particular condition without generating its counterpart. Intriguingly enough, when synthesizing new data it was observed that each feature could exclusively yield just a single status.

Decision Tree

To evaluate the model let's check our accuracy and classification report again.

```
Accuracy: 0.80
Classification Report:
              precision    recall  f1-score   support

     0         0.51         0.06         0.11         2984
     1         0.81         0.98         0.89        11742

 accuracy          0.80          0.80          0.80        14726
  macro avg         0.66         0.52         0.50        14726
 weighted avg         0.75         0.80         0.73        14726
```

Figure 27

The model has an accuracy of 80. The Precision of 0 is 0.51 and 1 is 0.81. The model has a higher performance for the "ON" class compared to the "OFF" class. This is probably due to the dataset having a low sample of the "OFF" data. The weighted average F1-score is approximately 0.73, indicating that there is room for improvement.

A plot for the confusion matrix is generated for the model.

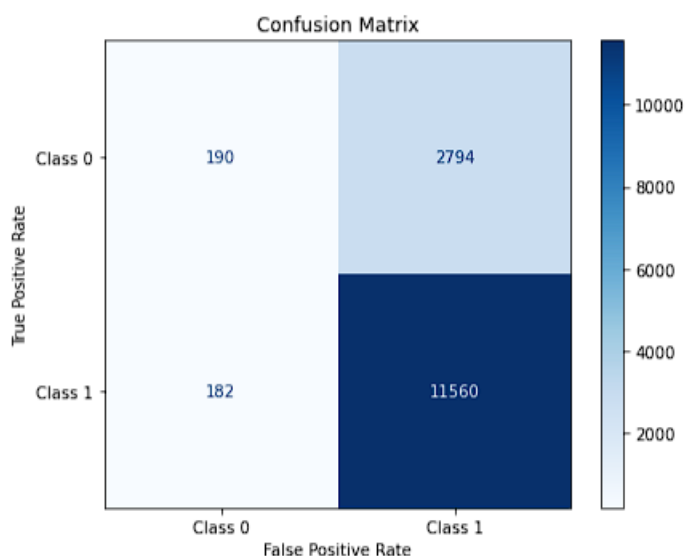


Figure 28

True Positives (TP): 190. This represents the number of instances that were actually "OFF" and were correctly predicted as "OFF" by the model. The model correctly identified 190 actions that resulted in "OFF".

False Negatives (FN): 2794 This represents the number of instances that were actually "OFF" but were incorrectly predicted as "ON" by the model. The model incorrectly identified 2794 actions as "ON" when they were "OFF".

False Positives (FP): 182. This represents the number of instances that were actually "ON" but were incorrectly predicted as "OFF" by the model. The model incorrectly identified 182 actions as "OFF" when they were actually "ON".

True Negatives (TN): 11560 This represents the number of instances that were actually "ON" and were correctly predicted as "ON" by the model. The model correctly identified 11560 actions that resulted in "ON".

Then a plot for the ROC curve is generated for the model.

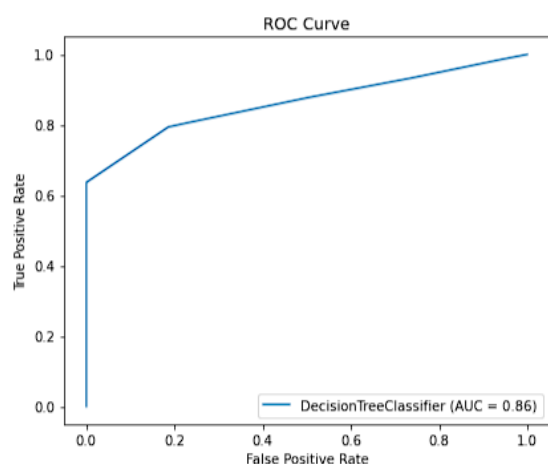


Figure 29

The more closely the curve adheres to the left and upper borders of the ROC space, the higher its precision. It can be observed that while initially hugging the left boundary, after a certain point it veers rightward indicating an initial high accuracy which subsequently deteriorates. Any points located beneath the diagonal line are indicative of classifiers with performance inferior to random conjecture; however, no such instances are visible here. The area under this ROC curve (AUC) serves as a gauge for classifier efficacy - where 1.0 signifies flawless classification and 0.5 denotes randomness equivalent to prediction quality. In this case, an AUC value of .86 suggests commendable classifier proficiency. The most balanced point on this curve would likely be situated around (.0,.65), wherein TPR exhibits maximal values relative to FPR.

Then a plot of histograms is generated for each feature and its true and generated states to see how much it was able to predict correctly.

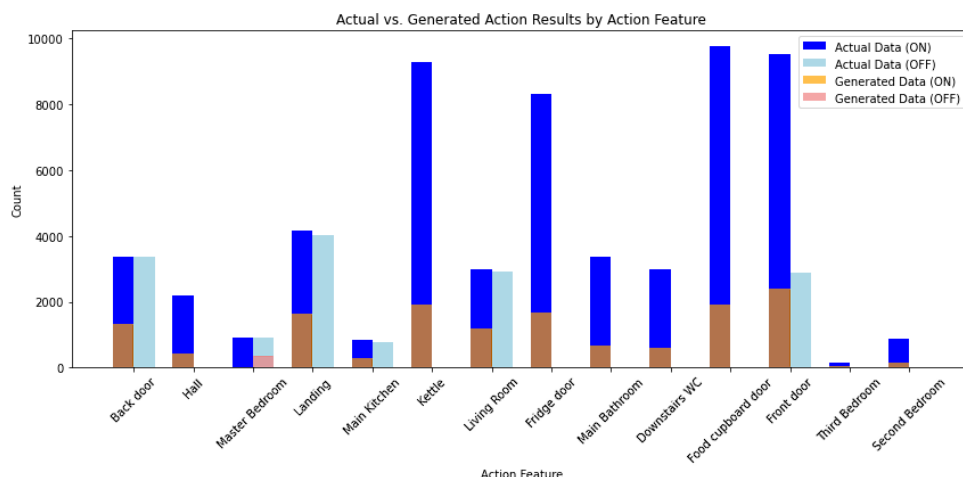


Figure 30

Here we can see that most of the states for each feature are predicted. But we can see some unique insights from it. We see that the generated data hardly consists of the OFF state. So almost none of the feature's actual OFF data were predicted. Only the Master Bedroom's OFF data was predicted but it couldn't predict the ON state for the Master Bedroom. But other than that most of the state were predicted.

When determining the most effective model, it is imperative to take into account the unique circumstances and demands of the task in question. In this case, a comparison between the Hidden Markov Model (HMM), Naive Bayes model, and Decision Tree model shows that based on their respective performances, the Decision Tree emerges as superior due to its highest accuracy rate among all three models. Accuracy serves as an important measure when classes are evenly distributed; however, if there's an imbalance within the dataset then relying solely on accuracy might not paint a complete picture.

Precision holds significance where False Positives carry high costs while Recall becomes vital when dealing with high costs associated with False Negatives. The F1-Score offers an equilibrium between Precision and Recall metrics. As per these measures provided here, it appears that for the "ON" class specifically - our Decision tree provides better balance compared to others.

The suggestion of potentially low sample data for the "OFF" class could indicate imbalanced classification which can affect how well our chosen model generalizes unseen data, especially concerning minority classes. Regardless of this potential imbalance though, the decision tree has exhibited relatively improved performance towards the 'OFF' category than HMM or Naïve Bayes models have been able to demonstrate.

Decision Trees come bearing advantages such as intuitiveness and interpretability along with handling non-linear relationships effectively but they do run the risk of overfitting particularly in cases involving deep trees. This was checked against the possibility of overfitting by comparing training accuracy to the overall model's accuracy revealing no signs indicative of Overfitting.

However, all models indicate the potential for improvement. By addressing the class imbalance, enhancing features, fine-tuning hyperparameters, and investigating ensemble methods performance can be optimized. The superior results of the Decision Tree may be attributed to their scalability and adaptability across a variety of data types and distributions. Given more intricate information or complex data sets other models might demonstrate higher efficacy.

The comprehensive analysis highlights the Decision Tree as an optimal choice due to its high accuracy rate, balanced precision-recall ratio, and ability to handle class imbalances effectively. Its interpretative capabilities along with flexibility further strengthen its suitability for this task. However context-specific requirements such as managing sequential data or emphasizing interpretation could influence model selection decisions differently.. Addressing these identified limitations while exploring ensemble techniques can refine our approach towards selecting a suitable model thereby optimizing performance tailored specifically for the given task.

Limitations

The Hidden Markov Model (HMM) is a potent statistical instrument for formulating sequences of observable data. Although it has been effectively employed in diverse areas such as voice recognition, finance, and bioinformatics, it possesses its own set of constraints, especially when handling intricate real-world situations like the one we addressed. The Naive Bayes classifier represents another machine learning model that has gained popularity due to its simplicity and efficacy. Originating from the application of Bayes' theorem, this model finds extensive use across various sectors including text categorization, spam identification, and sentiment analysis among others. Nevertheless, it also exhibits certain inherent drawbacks, particularly while dealing with sophisticated real-world datasets which were evident during our examination.

Decision trees provide intuitive comprehensibility along with an ability to formulate non-linear relationships but they are not devoid of their unique challenges and restrictions. It's imperative to consider these factors before choosing a decision tree for any particular assignment. In numerous practical scenarios, ensemble techniques like Random Forests or gradient-boosted trees have proved beneficial. These methods amalgamate outputs derived from multiple decision trees thereby overcoming some limitations associated with them resulting in enhanced predictive efficiency.

Limitations of the Data Used in the Analysis

The dataset's representation of all possible situations and actions may be incomplete, depending on the quantity of records available. Models that are trained based on limited data might become excessively attuned to this particular dataset, thereby struggling when it comes to generalizing unobserved scenarios. The dataset includes timestamps for each action undertaken. If there exist

inherent time-based patterns (such as daily or seasonal trends), models that do not factor in these temporal dynamics may fail at capturing such patterns accurately.

While the type of action taken and its outcome are included within the dataset, additional contextual information that could potentially enhance predictive performance is missing. This could include details about what instigated a specific action, external conditions during said action, or even previous sequences of actions - providing a more comprehensive understanding.

If due to certain aspects related to how data collection was carried out some types of actions or outcomes have higher chances of being recorded than others, this can introduce bias into any analysis performed using this data set. Without access to wider historical datasets it becomes difficult to compare current observed patterns against those from past periods; gaining insights into significant shifts in behavior over time or identifying anomalies requires an understanding of historical context.

This situation where one result "OFF" occurs much more frequently compared to "ON" poses challenges while modeling since class imbalance might lead model predictions towards the majority class thus reducing the accuracy of minority classes.

Limitations of the Hidden Markov Model (HMM) Used in the Analysis

A key presumption of Hidden Markov Models (HMM) is the Markovian property, which postulates that future states are solely dependent on present ones and not their preceding sequence. This constraint might result in overly simplified models incapable of encapsulating intricate dependencies within data. In our evaluation, we identified two concealed states: "active" and "inactive". However, this binary categorization could be too rudimentary for complex sequences influenced by various underlying factors. The determination of hidden state numbers is vital; an incorrect estimation can lead to overfitting or underfitting issues. Parameter estimation for HMMs presents a challenge particularly when dealing with large or sparse datasets like ours due to its computational intensity. Model training and parameter approximation necessitate iterative algorithms which can consume considerable time and resources - areas where our analysis was deficient. While HMM provides a structured method for sequence modeling, interpreting derived states and transitions may prove difficult if these do not have tangible real-world correlations – an issue potentially faced by our model as well.

Additionally, HMM presumes stationarity in the underlying process implying unchanging transition probabilities over time—an assumption often contradicted in practical scenarios where transition dynamics evolve constantly based on external influences uncaptured within the model framework—such as likelihood variations from 'inactive' to 'active'. There exists a potential risk wherein an HMM trained exclusively on specific data sets fails to generalize across unseen data, especially amidst significant alterations within inherent sequential dynamics—an obstacle encountered during our assessment.

Limitations of the Naive Bayes Model Used in the Analysis

The principal constraint is the presupposition of independent attributes. The model presumes that all features are entirely autonomous, given the class label. However, in numerous real-world circumstances, these characteristics can be interconnected. For instance, within our dataset, certain activities might correlate; however, the Naive Bayes model would consider them as separate entities possibly leading to misclassification errors. Regarding categorical data if any category does not appear during training but emerges in testing data then this could cause false predictions due to its probability being assigned zero by the model - something we encountered with OFF forecasts for our analysis.

In addition to this limitation is an assumption made by the Gaussian Naive Bayes variant which considers continuous features to follow a normal distribution and when it doesn't comply with this presumption it may lead to inaccurate predictions.

While simplicity often serves as an advantage especially when dealing with datasets having large feature counts there's also a potential downside where simplification becomes excessive particularly when handling complex data featuring intricate interrelationships among variables resulting in subpar predictive performance compared to more advanced models.

Although Naive Bayes provides classification probabilities understanding why certain features influence those probabilities greater than others proves difficult because it treats every attribute equally important regardless of their relevance or redundancy hence introducing noise into predictions at times. Our problem was binary ("on" vs "off") and while naive Bayes performs well on such problems had multiple classes been involved extension towards a better modeling approach might have proved beneficial considering the complexity of dependencies between sequences actions results present within our dataset suggesting suitability superior sophisticated models over simpler ones like ours

Limitations of the Decision Tree Model Used in the Analysis

A primary drawback of decision trees, particularly those that are deep, is their proclivity to overfitting. Although in this specific instance, the model demonstrated similar training and testing accuracies, there are many situations where without proper parameter adjustment or constraints, decision trees can grow exceedingly deep – conforming even to minor variations within the training data. Decision trees have a propensity for instability; slight alterations in the dataset could result in significant changes to tree structure — an issue amplified if the tree's depth and complexity are high. For example, introducing new data points or eliminating existing ones might lead our model to generate a substantially different tree which would affect its ability to make decisions.

Decision Trees may display a bias toward features with more levels or categories because they provide additional opportunities for splitting data - even when such splits do not offer much useful

information. About our current model: if some action-based features had significantly more unique values than others it could disproportionately impact both the structure of our model and its decision-making process.

While being powerful tools capable of capturing complex patterns, Decision Trees may sometimes lack the expressiveness required for certain relationships, especially those necessitating interactions between multiple features. The division boundaries formed by these models always run perpendicular to the feature axis which does not necessarily reflect the true relationship inherent within datasets accurately every time.

For cases wherein intricate or curved division boundary exists within datasets, we might find other models better suited than Decision Tree algorithms.

Decisions made by these algorithms at each step aim only at achieving the best split rather than considering whether said split will yield the lowest possible impurity several stages later- leading to occasionally sub-optimal results.

Although continuous variables can be handled using them doing so requires conversion into categorical counterparts based on set thresholds potentially resulting loss of crucial information should a nuanced correlation exist between the target variable & aforementioned continuous one.

Furthermore, unbalanced datasets pose a challenge due to imbalance favoring the majority class thereby affecting prediction accuracy-related minority classes negatively despite interpretability intuitiveness offered alongside modeling non-linear relationships. Thus, decision trees, while highly useful, are not without their own set of challenges and limitations.

Future Works that could be performed for Analysis of Finding Patterns

- Enrich the dataset with supplementary attributes that offer contextual insights. This may encompass external environmental factors, user demographics, or event catalysts. These elements can deliver a more comprehensive context for actions and assist in revealing subtle patterns.
- Considering the timestamp data, delve further into chronological analysis to discern time-bound patterns, cyclicity, or trends.
- Employ unsupervised learning methodologies to group similar activities or temporal periods. This could help highlight unique patterns or segments within the data that might not be immediately apparent through supervised techniques.
- Upon pinpointing unusual sequences of actions deviating from standard behavior; consider models capable of incorporating such findings into their analyses without eliminating them.
- Aim to enhance both quantity and diversity of data by amassing it over extended durations or from varying sources. Increased amounts of information can foster stronger and widely applicable pattern recognition capabilities.

- Trial advanced machine learning methods like LSTM (Long Short-Term Memory) or Transformer models as these are adept at capturing intricate relationships and sequences present in datasets.
- Meld different modeling approaches' strengths - for instance, HMM (Hidden Markov Model) combined with neural networks - this would boost pattern detection abilities significantly.
- Create interactive tools/dashboards enabling users to investigate potential correlations filter out irrelevant information, simulate possible scenarios, etc., thereby providing actionable intelligence supporting informed decision-making processes.
- Capture real-time action outcomes feeding this back into your model ensuring its continuous enhancement refinement.
- Integrate ancillary databases e.g., meteorological records and user behavior analytics offering additional perspective on improving pattern identification.
- Regularly compare the selected model's performance against other cutting-edge techniques guaranteeing optimal results and highlighting areas requiring improvement.

In summary, while current analytical endeavors provide valuable understanding regarding various actions along with associated outcomes there remains significant scope for future studies aimed at enhancing accuracy depth utility Adopting multiple strategies ensures holistic comprehension facilitating well-informed decisions.

Conclusion

The analytical endeavors undertaken in this report provide an in-depth look into the evaluation and effectiveness of various predictive models. These models, designed to understand and predict the outcomes of specific actions, offer both promising results and areas for further refinement.

A crucial part of this assessment was centered on the Decision Tree model. This model demonstrated an impressive accuracy rate of 80%, distinguishing it as a dependable option for the dataset being analyzed. When examining its precision metrics in more depth, we see a complex performance: one class has a precision score of 0.51 while another - 'ON' class - scores at 0.81. Such differences are critical when contemplating how this model might be applied in real-world situations; even though it performs exceptionally well in some areas, there is still potential for further enhancement, particularly regarding certain classes.

The study also drew attention to surprising inconsistencies found within specific features; instances were noted where these produced states that did not exist in actual datasets whilst failing to generate others that should have been included based on available data sources. The presence

of such anomalies emphasizes why ongoing evaluations are essential especially when working with varied and changing datasets – achieving high levels of accuracy isn't enough if predictions do not align with realistic scenarios or introduce unexpected new data points instead.

Although these observed discrepancies may lead some individuals towards dismissing entire analytic procedures altogether due to their concerning nature, doing so would overlook valuable insights they provide about weaknesses needing strengthening within current modeling techniques. Identifying those vulnerable aspects allows researchers/data scientists to focus their efforts more effectively thereby ensuring future versions become stronger and reliable over time.

This report's analytic endeavors provide an all-encompassing investigation into the assessment and effectiveness of different predictive models. These models, crafted to understand and predict the outcomes of particular actions, yield promising results as well as areas for further enhancement.

The report goes beyond merely pinpointing deficiencies. It extends its scope by offering a range of suggestions aimed at improving the analytical methodology and boosting model performance. The proposal to incorporate clustering techniques is particularly significant. Clustering inherently enables the recognition of distinct patterns and clusters within data sets which can be crucial in revealing concealed relationships or trends that may not be immediately discernible via conventional supervised methods. In today's rapidly progressing field of data analytics, adhering strictly to one method or model could prove restrictive. This limitation is acknowledged in this report through its endorsement of merging various modeling approaches. For example, combining the Hidden Markov Model (HMM) with neural networks has immense potential these hybrid methodologies leverage individual strengths from each model creating a prediction system that's both potent and intricate. Another progressive suggestion involves embracing cutting-edge machine learning practices like LSTM (Long Short-Term Memory) and Transformer models due to their structural design; these are proficient at comprehending sequences along with complex relationships present within datasets effectively predicting them too. As we deal with increasingly complicated data structures traditional modes might fall short thus adopting advanced techniques will place our analysis on par with leading-edge predictive analytics systems.

In summation, this report serves as an indication of the complexities and possibilities inherent in predictive analytics. While the current models and methodologies provide a solid foundation, the journey towards perfecting them is ongoing. The recommendations provided pave the way for a future where the analytical processes are not just accurate but also adaptable, comprehensive, and user-friendly. By embracing a multi-faceted approach, we can ensure that the insights derived are both deep and actionable, setting the stage for well-informed decisions in the ever-evolving domain of data science.

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Research Ethics Screening Form for Students

Middlesex University is concerned with protecting the rights, health, safety, dignity, and privacy of its research participants. It is also concerned with protecting the health, safety, rights, and academic freedom of its students and with safeguarding its own reputation for conducting high quality, ethical research.

This Research Ethics Screening Form will enable students to self-assess and determine whether the research requires ethical review and approval via the Middlesex Online Research Ethics (MORE) form before commencing the study. Supervisors must approve this form after consultation with students.

Student Name:		Email:
	Research project title:	
	Programme of study/module:	
Supervisor Name:		Email:

<i>Please answer whether your research/study involves any of the following given below:</i>		
1. ^H ANIMALS or animal parts.	☐ Yes	☒ No
2. ^M CELL LINES (established and commercially available cells - biological research).	☐ Yes	☒ No
3. ^H CELL CULTURE (Primary: from animal/human cells- biological research).	☐ Yes	☒ No
4. ^H CLINICAL Audits or Assessments (e.g. in medical settings).	☐ Yes	☒ No
5. ^X CONFLICT of INTEREST or lack of IMPARTIALITY. If unsure see "Code of Practice for Research" (Sec 3.5) at: https://unihub.mdx.ac.uk/study/spotlights/types/research-at-middlesex/research-ethics	☐ Yes	☒ No
6. ^X DATA to be used that is not freely available (e.g. secondary data needing permission for access or use).	☐ Yes	☒ No
7. ^X DAMAGE (e.g., to precious artefacts or to the environment) or present a significant risk to society).	☐ Yes	☒ No
8. ^X EXTERNAL ORGANISATION – research carried out within an external organisation or your research is commissioned by a government (or government body).	☐ Yes	☒ No
9. ^M FIELDWORK (e.g biological research, ethnography studies).	☐ Yes	☒ No
10. ^H GENETICALLY MODIFIED ORGANISMS (GMOs) (biological research).	☐ Yes	☒ No
11. ^H GENE THERAPY including DNA sequenced data (biological research).	☐ Yes	☒ No
12. ^M HUMAN PARTICIPANTS – ANONYMOUS Questionnaires (participants not identified or identifiable).	☐ Yes	☒ No
13. ^X HUMAN PARTICIPANTS – IDENTIFIABLE (participants are identified or can be identified): survey questionnaire/ INTERVIEWS / focus groups / experiments / observation studies.	☐ Yes	☒ No

14. ^H HUMAN TISSUE (e.g., human relevant material, e.g., blood, saliva, urine, breast milk, faecal material).	☐ Yes	☒ No
15. ^H ILLEGAL/HARMFUL activities research (e.g., development of technology intended to be used in an illegal/harmful context or to breach security systems, searching the internet for information on highly sensitive topics such as child and extreme pornography, terrorism, use of the DARK WEB, research harmful to national security).	☐ Yes	☒ No
16. ^X PERMISSION is required to access premises or research participants.	☐ Yes	☒ No
17. ^X PERSONAL DATA PROCESSING (Any activity with data that can directly or indirectly identify a living person). For example data gathered from interviews, databases, digital devices such as mobile phones, social media or internet platforms or apps with or without individuals'/owners' knowledge or consent, and/or could lead to individuals/owners being IDENTIFIED or SPECIAL CATEGORY DATA (GDPR) or CRIMINAL OFFENCE DATA.	☐ Yes	☒ No
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18. ^H RISK OF PHYSICAL OR PSYCHOLOGICAL HARM (e.g., TRAVEL to dangerous places in your own country or in a foreign country (see https://www.gov.uk/foreign-travel-advice), research with NGOs/humanitarian groups in conflict/dangerous zones, development of technology/agent/chemical that may be harmful to others, any other foreseeable dangerous risks).	☐ Yes	☒ No
19. ^X SECURITY CLEARANCE – required for research.	☐ Yes	☒ No
20. ^X SENSITIVE TOPICS (e.g., anything deeply personal and distressing, taboo, intrusive, stigmatising, sexual in nature, potentially dangerous, etc).	☐ Yes	☒ No

M – Minimal Risk; X – More than Minimal Risk. H – High Risk

If you have answered 'Yes' to ANY of the above questions, your application **REQUIRES** ethical review and approval using the MOREform **BEFORE commencing your research**. Please apply for approval using the MOREform (<https://moreform.mdx.ac.uk/>). Further guidance on making an application using the MOREform can be found at: www.tiny.cc/mdx-ethics.

If you have answered 'No' to ALL of the above questions, your application is Low Risk and you may NOT require ethical review and approval using the MOREform before commencing your research. Your research supervisor will confirm this below.

Student Signature:..... Date:.....

To be completed by the supervisor:

<i>Based on the details provided in the self-assessment form, I confirm that:</i>	Insert Y or N
The study is Low Risk and <i>does not require</i> ethical review & approval using the MOREform	
The study <i>requires</i> ethical review and approval using the MOREform.	

Supervisor Signature:..... Date:.....